BEGINNER'S GUIDE

Large Language Models (LLMs)

From Zero to AI Hero — Everything You Need to Know

15 Slides • End-to-End Coverage • Beginner Friendly

1 What is an LLM?

A Large Language Model (LLM) is an AI system trained on massive amounts of text to understand and generate human language — just like how YOU learned to speak by reading, listening, and practicing!

Learns from Text

Trained on books, websites, code, and articles — billions of words!

Predicts Words

It predicts the NEXT best word, one at a time, to form a response.

Generates Language

Answers questions, writes essays, translates, codes, and more.

Example: You ask "What is gravity?" → LLM reads your question → predicts the best response → you get a clear explanation!

2 A Brief History of LLMs

2017

Transformer

Google invents the Transformer architecture — the engine behind all modern LLMs

2018

BERT & GPT-1

First generation of large pre-trained models appear

2020

GPT-3

OpenAI releases GPT-3 with 175B parameters. World is amazed!

2022

ChatGPT

ChatGPT launches. 1 million users in 5 days!

2023+

LLM Boom

Gemini, Claude, Llama, Mistral & many more launch globally

Key Insight: LLMs grew from research projects to products used by 500+ million people worldwide in just 6 years!

3 How LLMs Work — The Transformer

INPUT
Your Prompt

TOKENIZE
Break into word pieces

ATTENTION
Understand Context

PREDICT
Next Token Prob.

OUTPUT
Generated Text

Attention — The Secret Sauce

Sentence: "The bank was steep, so she slipped on the **bank**"

Attention helps the model know the 2nd "bank" means RIVER BANK (not a financial bank) by looking at ALL words together — like reading the full sentence before answering.

Tokenization Example

"unbelievable" →

un	believ	able

Words split into small chunks → numbers

4 Training — How LLMs Learn

Phase 1

Pre-Training

Feed TRILLIONS of words from the internet, books, code.
Model learns grammar, facts, reasoning by predicting next words.

Trillions of tokens
Training Data

Phase 2

Fine-Tuning

Train on curated, high-quality examples.
Model learns SPECIFIC tasks: answering, summarizing, coding.

Millions of
Example Pairs

Phase 3

RLHF

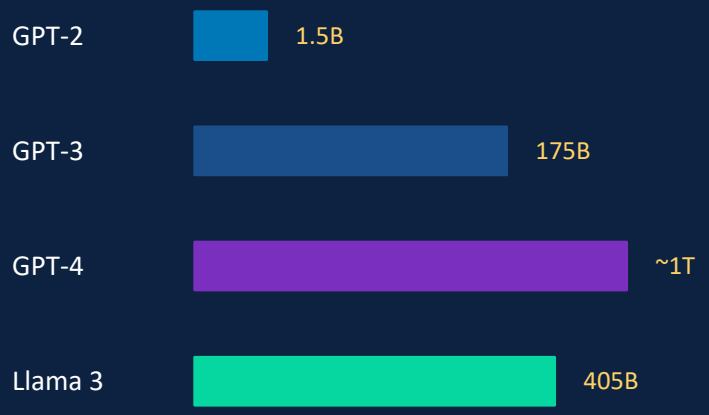
Humans rate responses. Model learns to be helpful, honest, and safe using Reinforcement Learning from Human Feedback.

Human
Feedback Loop

5 Key Concepts — Parameters & Tokens

Parameters (Weights)

Think of parameters as the model's "memories" — millions/billions of numbers that store everything it learned.

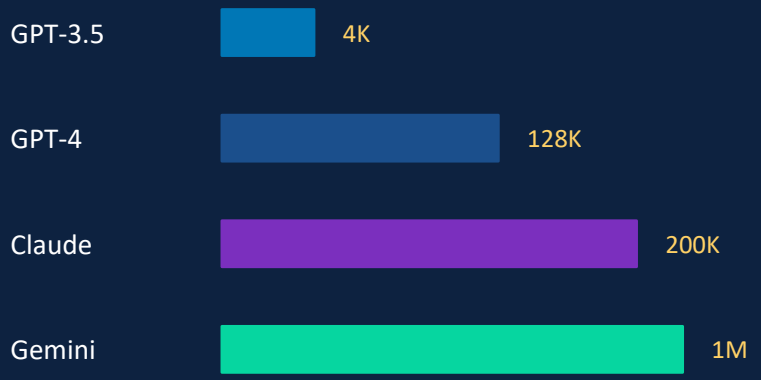


More parameters = More knowledge (but needs more computing power!)

Tokens — The Building Blocks

Tokens are pieces of words. "Hello world!" = 3 tokens. Models have a "context window" — how many tokens they can see at once.

Context Window Sizes:



Bigger context = reads longer documents, remembers more in conversation!

6 Popular LLMs — Who's Who?

ChatGPT (GPT-4o)

OpenAI

General purpose, coding, image understanding

Closed Source

Claude (Sonnet/Opus)

Anthropic

Long docs, safety, nuanced writing

Closed Source

Gemini (Ultra/Pro)

Google

Search integration, multimodal, long context

Closed Source

Llama 3 (Meta)

Meta AI

Free to use, self-host, customizable

Open Source

Mistral (7B / 8x7B)

Mistral AI

Lightweight, fast, efficient, open weights

Open Source

7 Prompt Engineering — Talk to AI Effectively

❌ Weak Prompt

"Write something about marketing"

❓ Too vague — AI doesn't know your goal, audience, tone, format, or length.

✅ Strong Prompt

"Write a 200-word Instagram caption for a coffee shop targeting students, using a friendly tone with 3 hashtags."

✅ Specific role + length + audience + tone + format!

5 Prompting Techniques Every Beginner Should Know:

Give a Role

Say "Act as a doctor" or "You are a Python expert"

Few-Shot Examples

Show 2-3 examples of what you want before your request

Step by Step

Add "Think step by step" for complex reasoning tasks

Specify Format

Ask for bullet points, table, JSON, or a specific word count

Iterate

Refine! If unsatisfied, say what to improve and try again

What Can LLMs Do? — Capabilities

Writing & Editing

Blog posts, emails, proofreading, summarizing documents

Coding

Write, debug, explain code in Python, JS, SQL, and 50+ languages

Translation

Translate between 100+ languages instantly with cultural nuance

Teaching & Tutoring

Explain complex topics simply, answer questions, create quizzes

Research & Analysis

Summarize papers, extract insights, compare information

Creative Work

Stories, poems, scripts, brainstorming, world-building

Data & Math

Solve equations, analyze data, explain statistics step-by-step

Automation & Agents

Automate tasks, browse web, use tools, run multi-step workflows

9 Limitations & Hallucinations — Know the Risks

Hallucinations

LLMs sometimes confidently make up false facts, fake citations, or wrong answers. Always verify important info!

✅ Cross-check with reliable sources

Knowledge Cutoff

Models are trained on data up to a certain date. They don't know about recent events unless they can browse the web.

✅ Use web-enabled versions for current events

Math & Logic Errors

LLMs struggle with precise arithmetic and multi-step logical reasoning, especially without tools.

✅ Use code interpreter or calculators

Privacy Risks

Anything you type may be used for training. Never share passwords, medical records, or personal data!

✅ Use private/enterprise versions for sensitive data

Bias

Trained on human-written text, LLMs can reflect societal biases. Outputs may be unfair or stereotypical.

✅ Critically evaluate outputs, ask for balanced perspectives

Context Limits

Even with big context windows, very long conversations can cause the model to 'forget' early instructions.

✅ Re-state key context for long sessions

RAG — Connecting LLMs to Your Data

📚 Analogy: RAG is like giving an LLM a textbook + Google search for YOUR specific documents. Instead of relying only on training data, it retrieves relevant info first, then answers!

User Question

"What is our refund policy?"

→

Search Vector DB

Find relevant docs from your knowledge base

→

Retrieve Context

Get the 3 most relevant document chunks

→

LLM Combines

Merges question + retrieved docs into one prompt

→

Accurate Answer

Response grounded in your actual documents!

Why RAG? LLMs alone can't access your private PDFs, databases, or latest info. RAG solves this!

- 🏢 Customer Support Bot that reads your FAQs
- 📋 Legal AI that cites actual case documents
- 🏥 Medical chatbot grounded in clinical guidelines
- 💼 Company knowledge base assistant

LLM Agents — AI That Takes Action

🤖 An LLM Agent = LLM Brain + Tools + Memory + Goal. Instead of just chatting, it DOES things: searches the web, runs code, sends emails, and plans multi-step tasks autonomously!

The Agent Loop:

1. PERCEIVE
Receive Goal

→

2. THINK
Plan Steps

→

3. ACT
Use a Tool

→

4. OBSERVE
Check Result

→

5. REPEAT
or Finish

Tools Agents Can Use:

- Web Search
- Code Runner
- Email / Slack
- File Read/Write
- Database Query
- API Calls
- Calculator
- Calendar

Real-World Use Cases — LLMs in Action

Healthcare

- Medical report summarization
- Drug interaction checks
- Patient Q&A chatbots
- Clinical trial data extraction

Legal

- Contract review & drafting
- Case research assistant
- Regulatory compliance checks
- Legal document translation

Education

- Personalized tutoring bots
- Essay feedback & grading
- Curriculum design
- Language learning practice

E-Commerce

- Product description writing
- Customer support chatbots
- Personalized recommendations
- Review summarization

Finance

- Financial report analysis
- Fraud detection narrative
- Earnings call summarization
- Risk assessment reports

Software Dev

- Code generation & review
- Bug detection & fixing
- Documentation writing
- Test case generation

Your AI Learning Roadmap — Getting Started

Month 1–2

Foundations

- ✓ Use ChatGPT / Claude daily
- ✓ Learn prompting basics
- ✓ Take a free AI course (fast.ai, Coursera)
- ✓ Understand what tokens & parameters are

Month 3–4

Build Skills

- ✓ Learn Python basics
- ✓ Call OpenAI/Anthropic API
- ✓ Build a simple chatbot
- ✓ Explore LangChain or LlamaIndex

Month 5–6

Projects

- ✓ Build a RAG app on your own docs
- ✓ Fine-tune a small model (Hugging Face)
- ✓ Deploy to production (AWS/GCP/Azure)
- ✓ Share on GitHub & LinkedIn

Popular AI Career Roles:

Prompt Engineer	ML Engineer	AI Product Manager	LLM App Developer	AI Researcher

Ethics, Safety & The Future of AI

Misinformation

LLMs can generate convincing false content at scale. Critical thinking is essential.

Data Privacy

Who owns the data used to train? Users must understand consent and usage policies.

Bias & Fairness

Biased training data → biased outputs. Diverse data and audits are crucial.

Environmental Impact

Training large models uses enormous energy. Efficiency matters for sustainability.

What's Next — The Future of LLMs

Multimodal Models	See, hear, and speak — not just text (images, audio, video)
Reasoning Models	Think step-by-step like humans (OpenAI o1, DeepSeek R1)
Autonomous Agents	AI that plans, acts, and completes complex goals on its own
AI Scientists	LLMs designing experiments, discovering drugs, solving science

"The goal of AI is not to replace humans, but to amplify what we can achieve together." — Start learning today!

Quick Glossary — Key AI Terms

LLM	Large Language Model — AI trained on massive text data to understand & generate language
Parameters	The numerical 'weights' in a model — billions of numbers that encode learned knowledge
Temperature	Controls randomness of output (0 = deterministic/factual, 1+ = creative/varied)
RAG	Retrieval-Augmented Generation — LLM searches your documents to give grounded answers
Embedding	Converting text to numbers (vectors) so AI can understand similarity and meaning
Hallucination	When an LLM confidently states false information — always verify important facts!
Token	A chunk of text (word piece) that the model processes; ~4 chars or ¾ of a word on average
Prompt	The input you give to an LLM; quality of prompt greatly affects quality of output
Context Window	Max tokens an LLM can process at once — determines how much it can 'remember' per session
Fine-tuning	Training an existing model further on custom data to specialize it for your task
Agent	An LLM that uses tools (web, code, APIs) to autonomously complete multi-step tasks
RLHF	Reinforcement Learning from Human Feedback — training method to make LLMs helpful & safe

🚀 Keep Learning: fast.ai | Hugging Face | DeepLearning.AI | Andrej Karpathy YouTube | Papers With Code